

Assignment 3A Global Baseline Estimate

AUTHOR

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LIBRARIES

```
library(readxl)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)
library(tidyr)

## Importing my movie survey data

movie_data <- read_excel("C:\\Users\\kingk\\OneDrive\\Documents\\CUNY MASTERS PROGRAM-DATA SCIENCE\\movie_data.xlsx")

movie_data_long <- movie_data %>%
  pivot_longer(
    cols = Vikings:`Game of Thrones`,
    names_to = "title",
    values_to = "rating",
    values_transform = list(rating = as.character)
  )

movie_data_long <- movie_data_long %>%
  mutate(
    rating = as.numeric(rating)
  )
```

Warning: There was 1 warning in `mutate()`.
i In argument: `rating = as.numeric(rating)`.
Caused by warning:
! NAs introduced by coercion

```
glimpse(movie_data_long)
```

Rows: 80

Columns: 5

```
$ Timestamp      <dtm> 2026-09-19 10:48:50, 2026-09-19 10:48:50, 2026-09-19 ...
$ `Respondent ID` <dbl> 1001, 1001, 1001, 1001, 1001, 1001, 1001, 1001, 1001, ...
$ `First Name`    <chr> "Kingsley", "Kingsley", "Kingsley", "Kingsley", "Kings...
$ title           <chr> "Vikings", "See", "Vampire Dairies", "Beauty in Black"...
$ rating          <dbl> 4, 3, 4, 5, 4, 5, 4, 5, 5, 4, 4, 3, 4, 4, 5, 4, NA, 4,...
```

```
file.choose()
```

```
[1] "C:\\Users\\kingk\\OneDrive\\Documents\\CUNY MASTERS PROGRAM-DATA SCIENCE\\Data Acquisition &
Management 607\\Assignments\\Assignment - Week 3\\3A\\MOVIE SURVEY - KINGSLEY KPOKU.xlsx"
```

LOOKING INTO THE DATA

```
names(movie_data)
```

```
[1] "Timestamp"      "Respondent ID"   "First Name"      "Vikings"
[5] "See"            "Vampire Dairies" "Beauty in Black" "Breaking Bad"
[9] "Black Panther"  "Supacell"        "Troy"            "Avatar"
[13] "Game of Thrones"
```

```
head(movie_data)
```

```
# A tibble: 6 × 13
  Timestamp      `Respondent ID` `First Name` Vikings See
  <dtm>          <dbl> <chr>         <chr>    <chr>
1 2026-09-19 10:48:50      1001 Kingsley    4.0      3.0
2 2026-09-19 10:51:54      1002 Bright      4.0      3.0
3 2026-09-19 10:53:55      1003 Skinner     Not Seen Not Seen
4 2026-09-19 10:58:16      1004 Onam Eyram   3.0      Not Seen
5 2026-09-19 11:09:15      1005 Juliet      3.0      4.0
6 2026-09-19 11:11:46      1006 Dylan       4.0      Not Seen
# i 8 more variables: `Vampire Dairies` <chr>, `Beauty in Black` <chr>,
#   `Breaking Bad` <chr>, `Black Panther` <dbl>, Supacell <chr>, Troy <chr>,
#   Avatar <dbl>, `Game of Thrones` <chr>
```

CLEANING COLUMN NAMES

```
names(movie_data) <- trimws(names(movie_data))
```

```
names(movie_data)
```

```
[1] "Timestamp"      "Respondent ID"  "First Name"    "Vikings"
[5] "See"            "Vampire Dairies" "Beauty in Black" "Breaking Bad"
[9] "Black Panther"  "Supacell"       "Troy"          "Avatar"
[13] "Game of Thrones"
```

CALCULATE THE GLOBAL MEAN

```
global_mean <- mean(movie_data_long$rating, na.rm = TRUE)
```

```
global_mean
```

```
[1] 4.16129
```

```
## the average movie rating is 4.161
```

CALCULATE THE AVERAGE RATING FOR EACH MOVIE

```
movie_average <- movie_data_long %>%
  group_by(title) %>%
  summarise(
    movie_avg = mean(rating, na.rm = TRUE)
  )
```

```
movie_average
```

```
# A tibble: 10 × 2
```

	title	movie_avg
	<chr>	<dbl>
1	Avatar	4.38
2	Beauty in Black	4.17
3	Black Panther	4.25
4	Breaking Bad	4.43
5	Game of Thrones	4.29
6	See	3.5
7	Supacell	4.4
8	Troy	4
9	Vampire Dairies	4
10	Vikings	3.83

CALCULATE THE MOVIE ADJUSTMENT

```
movie_average <- movie_average %>%
  mutate(
    movie_adjustment = round(movie_avg - global_mean, 2)
```

```

) %>%
  arrange(desc(movie_adjustment))

movie_average

```

```

# A tibble: 10 × 3
  title          movie_avg movie_adjustment
  <chr>          <dbl>         <dbl>
1 Breaking Bad    4.43            0.27
2 Supacell        4.4             0.24
3 Avatar          4.38            0.21
4 Game of Thrones 4.29            0.12
5 Black Panther   4.25            0.09
6 Beauty in Black 4.17            0.01
7 Troy            4              -0.16
8 Vampire Dairies 4                -0.16
9 Vikings         3.83           -0.33
10 See            3.5            -0.66

```

from the analysis below only black panther, avatar, game of thrones and troy have an average above

VISUALIZE - MOVIE ADJUSTMENT

```
library(ggplot2)
```

```

ggplot(movie_average, aes(
  x = reorder(title, movie_adjustment),
  y = movie_adjustment
)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Movie Adjustment from Global Mean",
    x = "Movie",
    y = "Movie Adjustment"
  ) +
  theme_minimal()

```

Movie Adjustment from Global Mean

